

Mathematics for Electrical Sciences

Complex Numbers · Vector Algebra · Integral Calculus · Differential Equations & Laplace Transforms

Complete handbook for the Kerala Diploma Revision 2026 curriculum. Covers all four modules with deep explanations, worked examples, interactive calculators, Malayalam support, and exam-oriented practice.

PROGRAMME	Biomedical Engg, ECE, EEE, EE, Electronics Engg, Embedded Systems, Electronics & Computer Engg, EV Technology, IC Design, Instrumentation, Micro Electronics, Renewable Energy
COURSE TYPE	Theory
CONTACT HOURS	60 (L:45 + T:15)
SELF-LEARNING	60 hours (suggested)
CIE / ESE	40 / 60

MODULE I	Complex Numbers (14 h)
MODULE II	Vector Algebra (15 h)
MODULE III	Integral Calculus (15 h)
MODULE IV	Differential Equations & Laplace Transforms (16 h)



Official Source Declaration

Source: Kerala Polytechnic Revision 2026 Syllabus — Course 2001B (Mathematics for Electrical Sciences). This handbook is an educational study aid developed for students. It is not an official publication of the examining body.

⚠ Verified Inconsistencies: The PDF shows "Mathematics for Electrical Sciences" as both the course title and the textbook author (SITTTR Kalamasser). This appears to be a formatting artifact — the textbook is likely the same SITTTR publication used for 2001A, with a different title. The assessment methodology table is garbled but has been interpreted consistently with the data provided. All other data reconcile.



Course Dashboard

60

CONTACT HOURS

4

CREDITS

40+60

CIE / ESE

4

MODULES

Prerequisites

Fundamentals of Engineering Mathematics I (Course 1002) — Trigonometry, Determinants, Differentiation.

Teaching & Assessment Scheme

Teaching Scheme	Self-Learning/Week	Total Hours	Credits	CIE	ESE
3:1:0:0 (L:T:P:R)	5.5 h	60	4	40	60

Module-wise Instructional Hours				
Module	Title	Lecture (L)	Tutorial (T)	Total
I	Complex Numbers	10	4	14
II	Vector Algebra	12	3	15
III	Integral Calculus	11	4	15
IV	Differential Equations & Laplace Transforms	12	4	16
Total			60	

How to Use This Handbook

1

Understand

Read each module's explanation, formulae, and worked examples.

2

Visualise

Study the SVG diagrams and interactive calculators.

3

Apply

Solve the module-wise questions and try the model paper.

4

Revise

Use the Quick Revision cards and glossary before exams.

Course Outcomes

CO1

Apply **concepts of complex numbers** to solve problems. (Bloom: Apply)

CO2

Identify and apply **vector algebra techniques** to solve problems involving scalar and vector quantities. (Bloom: Apply)

CO3

Use **integration techniques** to solve basic engineering problems. (Bloom: Apply)

CO4

Solve **first-order differential equations** and understand Laplace transforms. (Bloom: Apply)

CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

Legends: 1–Low; 2–Medium; 3–High; – No Correlation



Curriculum Coverage Map

Module → Topics → Hours → CO → Handbook Anchor

Module	Major Topics	Hours	CO	Section
I	Complex number system, operations, Argand plane, modulus, amplitude, polar form	14	CO1	Module 1
II	Scalars & vectors, addition, subtraction, scalar multiplication, dot/cross product, applications (area, work, moment)	15	CO2	Module 2
III	Integration as reverse, standard integrals, substitution, parts, definite integrals, area under curve	15	CO3	Module 3
IV	Differential equations (order/degree, variable separable, linear), Laplace transforms (elementary functions, shifting property, inverse)	16	CO4	Module 4



How the Modules Connect

Module 1

Complex Numbers

Foundation for AC circuit analysis, signal processing, and control systems in electrical engineering.

Module 2

Vector Algebra

Essential for electromagnetics, mechanics, and field analysis — force, torque, work, and moments.

Module 3

Integral Calculus

Area, volume, work, and modelling dynamic systems via accumulation and area under curves.

Module 4

D.E. & Laplace

Modelling electrical circuits, control systems, and signal processing using differential equations and transforms.

MODULE I

Complex Numbers

14 hours (L:10 + T:4)

CO1

Bloom: Apply

Learning Goals

- Understand the complex number system
- Perform fundamental operations
- Represent complex numbers graphically
- Find modulus, amplitude, and polar form
- Solve engineering problems

Key Facts

- $i = \sqrt{-1}$
- $z = a + ib$
- $|z| = \sqrt{a^2 + b^2}$
- $\arg(z) = \tan^{-1}(b/a)$
- Polar: $z = r(\cos \theta + i \sin \theta)$

The Complex Number System

A **complex number** is a number of the form $z = a + ib$, where **a** and **b** are real numbers, and **i** is the imaginary unit defined by $i^2 = -1$.

- **Real part:** $\text{Re}(z) = a$
- **Imaginary part:** $\text{Im}(z) = b$

$$z = a + ib \cdot a, b \in \mathbb{R} \cdot i^2 = -1$$

Example: $z = 3 + 4i$ has $\text{Re} = 3$ and $\text{Im} = 4$.

Malayalam support: കോംപ്ലക്സ് സംഖ്യകൾ (complex numbers) $a + ib$ രൂപത്തിൽ എഴുതാം. ഇവിടെ 'i' $\sqrt{-1}$ നെ സൂചിപ്പിക്കുന്നു. a യും b യും യഥാക്രമം റിയൽ ഭാഗവും ഇമിറ്ററി ഭാഗവുമാണ്.

 **Tip:** The real and imaginary parts are always real numbers. The imaginary part includes the coefficient of i , not i itself.

Conjugate and Equality

Conjugate: The conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

$$\bar{z} = a - ib$$

Equality: Two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal iff $a_1 = a_2$ **and** $b_1 = b_2$.

Example: If $2x + 3iy = 4 + 6i$, then $2x = 4 \Rightarrow x = 2$, and $3y = 6 \Rightarrow y = 2$.



Tip: The conjugate is used to rationalise denominators and find modulus.

Fundamental Operations

For $z_1 = a + ib$ and $z_2 = c + id$:

- **Addition:** $z_1 + z_2 = (a+c) + i(b+d)$
- **Subtraction:** $z_1 - z_2 = (a-c) + i(b-d)$
- **Multiplication:** $z_1 \cdot z_2 = (ac - bd) + i(ad + bc)$
- **Division:** $z_1 / z_2 = (a+ib)/(c+id) \times (c-id)/(c-id) = [(ac+bd) + i(bc-ad)] / (c^2 + d^2)$

Example (Multiplication): $(2+3i)(1-4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$

Example (Division): $(3+4i)/(1-2i) = (3+4i)(1+2i)/5 = (3+6i+4i+8i^2)/5 = (-5+10i)/5 = -1 + 2i$



Common mistake: In division, always multiply numerator and denominator by the conjugate of the denominator.

Graphical Representation — Argand Plane

Complex numbers are represented on the **Argand plane** (complex plane) with the real part on the horizontal axis and the imaginary part on the vertical axis.

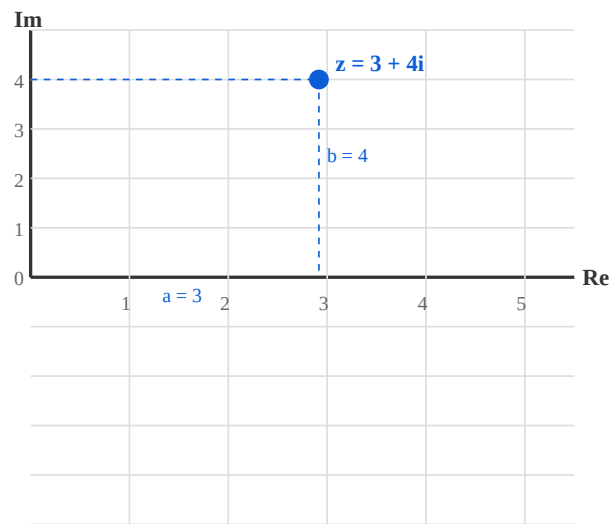


Figure 1: Argand plane showing $z = 3 + 4i$ with real part $a = 3$ and imaginary part $b = 4$.

Malayalam support: Argand plane- complex numbers- points represent . Horizontal axis real part-um vertical axis imaginary part-um .

Modulus and Amplitude

Modulus (Absolute value): $|z| = \sqrt{a^2 + b^2}$. It is the distance from the origin to the point (a, b) .

Amplitude (Argument): $\theta = \arg(z) = \tan^{-1}(b/a)$, measured in degrees (or radians).

$$|z| = \sqrt{a^2 + b^2} \quad \cdot \quad \theta = \tan^{-1}(b/a)$$

Example: For $z = 3 + 4i$: $|z| = \sqrt{9+16} = 5$, $\theta = \tan^{-1}(4/3) \approx 53.13^\circ$

! Common mistake: The argument depends on the quadrant. Always check the signs of a and b . In QII, $\theta = 180^\circ - \tan^{-1}(b/|a|)$.

Polar Form

A complex number can be expressed in **polar form** as:

$$z = r(\cos \theta + i \sin \theta) \cdot r = |z|, \theta = \arg(z)$$

This is also written as $z = r \angle \theta$ or $z = r \operatorname{cis} \theta$.

Example: For $z = 3 + 4i$: $r = 5$, $\theta = 53.13^\circ \Rightarrow z = 5(\cos 53.13^\circ + i \sin 53.13^\circ)$

 **Tip:** Polar form is especially useful for multiplication and division — multiply moduli and add arguments.

Solved Problems

Problem 1: Find the modulus and argument of $z = -1 + i$.

Solution: $a = -1$, $b = 1$. $r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$. $\theta = 180^\circ - \tan^{-1}(1/1) = 180^\circ - 45^\circ = 135^\circ$.

Problem 2: Express $z = 2 - 2i$ in polar form.

Solution: $r = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$. $\theta = -\tan^{-1}(2/2) = -45^\circ$. $z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ))$.

Problem 3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.

Solution: $(1+2i)(3-i) = 3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$.

 **Engineering Application:** Complex numbers model AC circuits — impedance $Z = R + iX$, with magnitude giving total opposition and phase angle giving phase shift.



Complex Number Calculator

Enter real and imaginary parts to compute modulus, argument, and polar form.

Real part (a)

Imaginary part (b)

3

4

$z = 3 + 4i$ | $|z| = 5.000$ | $\arg = 53.13^\circ$ | Polar: $5.000(\cos 53.13^\circ + i \sin 53.13^\circ)$

MODULE II

Vector Algebra

15 hours (L:12 + T:3)

CO2

Bloom: Apply



Learning Goals

- Distinguish scalars and vectors
- Perform addition, subtraction, scalar multiplication
- Compute dot and cross products
- Apply vectors to engineering problems: area, work, moment



Key Facts

- $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$
- $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \cdot \hat{\mathbf{n}}$
- $\text{Work} = \mathbf{F} \cdot \mathbf{d}$
- $\text{Moment} = \mathbf{r} \times \mathbf{F}$
- $\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}|$

Scalars, Vectors, and Unit Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, temperature, speed).
- **Vector:** A quantity with both magnitude and direction (e.g., velocity, force, displacement).

Unit vectors: $\hat{i}$, $\hat{j}$, $\hat{k}$ along x, y, z axes respectively. Each has magnitude 1.

Position vector: $\mathbf{r} = x \hat{i} + y \hat{j} + z \hat{k}$.

Malayalam support: Scalar-മാത്ര മാനിത (ഉദാ: mass). Vector-
മാത്ര മാനിത-ഉപദിഷ്ഠിത (ഉദാ: velocity). Unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$ axes-
മാത്ര മാനിത.

Magnitude and Unit Vector

Magnitude: $|\mathbf{v}| = \sqrt{x^2 + y^2 + z^2}$ for $\mathbf{v} = x \hat{i} + y \hat{j} + z \hat{k}$.

Unit vector: $\hat{v} = \mathbf{v} / |\mathbf{v}|$ (direction of $\mathbf{v}$).

Example: $\mathbf{v} = 3\hat{i} + 4\hat{j}$. $|\mathbf{v}| = 5$. $\hat{v} = (3/5)\hat{i} + (4/5)\hat{j}$.

Vector Addition, Subtraction, Scalar Multiplication

- **Addition:** $\mathbf{v} + \mathbf{w} = (v_1 + w_1)\hat{i} + (v_2 + w_2)\hat{j} + (v_3 + w_3)\hat{k}$
- **Subtraction:** $\mathbf{v} - \mathbf{w} = (v_1 - w_1)\hat{i} + (v_2 - w_2)\hat{j} + (v_3 - w_3)\hat{k}$
- **Scalar multiplication:** $c \cdot \mathbf{v} = (c v_1)\hat{i} + (c v_2)\hat{j} + (c v_3)\hat{k}$

Example: $\mathbf{v} = 2\hat{i} + 3\hat{j}$, $\mathbf{w} = 1\hat{i} - 2\hat{j}$. $\mathbf{v} + \mathbf{w} = 3\hat{i} + 1\hat{j}$, $\mathbf{v} - \mathbf{w} = 1\hat{i} + 5\hat{j}$, $3\mathbf{v} = 6\hat{i} + 9\hat{j}$.

Dot Product

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \cos \theta \quad \cdot \quad \mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$$

- **Angle:** $\cos \theta = (\mathbf{v} \cdot \mathbf{w}) / (|\mathbf{v}| |\mathbf{w}|)$
- **Perpendicular:** $\mathbf{v} \perp \mathbf{w} \Leftrightarrow \mathbf{v} \cdot \mathbf{w} = 0$
- **Work done:** $W = \mathbf{F} \cdot \mathbf{d}$ (force $\times$ displacement)

Example: $\mathbf{v} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$, $\mathbf{w} = 1\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$. $\mathbf{v} \cdot \mathbf{w} = 2(1) + 3(-2) = 2 - 6 = -4$.
 $|\mathbf{v}| = \sqrt{13}$, $|\mathbf{w}| = \sqrt{5}$, $\cos \theta = -4 / (\sqrt{13} \cdot \sqrt{5}) \approx -0.496$.

Cross Product

$$\mathbf{v} \times \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \sin \theta \cdot \hat{\mathbf{n}} \quad \cdot \quad \mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2) \hat{\mathbf{i}} - (v_1 w_3 - v_3 w_1) \hat{\mathbf{j}} + (v_1 w_2 - v_2 w_1) \hat{\mathbf{k}}$$

- **Parallel:** $\mathbf{v} \parallel \mathbf{w} \Leftrightarrow \mathbf{v} \times \mathbf{w} = \mathbf{0}$
- **Area of parallelogram:** $|\mathbf{v} \times \mathbf{w}|$
- **Area of triangle:** $\frac{1}{2} |\mathbf{v} \times \mathbf{w}|$
- **Moment (torque):** $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

Example: $\mathbf{v} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$, $\mathbf{w} = 1\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$. $\mathbf{v} \times \mathbf{w} = (3 \cdot 0 - 0 \cdot (-2)) \hat{\mathbf{i}} - (2 \cdot 0 - 0 \cdot 1) \hat{\mathbf{j}} + (2 \cdot (-2) - 3 \cdot 1) \hat{\mathbf{k}} = -7\hat{\mathbf{k}}$.

Applications: Area, Work, Moment

Area of parallelogram: Vectors $a = 2\hat{i}+3\hat{j}$, $b = 1\hat{i}-2\hat{j}$. $|a \times b| = |-7\hat{k}| = 7$ sq. units.

Area of triangle: $\frac{1}{2} \times 7 = 3.5$ sq. units.

Work done: Force $F = 3\hat{i}+4\hat{j}$ N, displacement $d = 2\hat{i}+1\hat{j}$ m. $W = F \cdot d = 3(2)+4(1) = 10$ J.

Moment: Position $r = 2\hat{i}+3\hat{j}$, Force $F = 1\hat{i}-2\hat{j}$. $\tau = r \times F = (3 \cdot 0 - 0 \cdot (-2))\hat{i} - (2 \cdot 0 - 0 \cdot 1)\hat{j} + (2 \cdot (-2) - 3 \cdot 1)\hat{k} = -7\hat{k}$ N·m.

 **Engineering Application:** Vectors are essential in electromagnetics (electric/magnetic fields), mechanics (forces, torques), and structural analysis.

Vector Calculator

Enter components of two vectors (a, b) and compute dot product, cross product, magnitudes, angle, area, and work (with force and displacement).

a_1

2

a_2

3

a_3

0

b_1

1

b_2

-2

b_3

0

$a \cdot b = -4.000$ | $a \times b = (0.000, 0.000, -7.000)$ | $|a| = 3.606$ | $|b| = 2.236$ | $\theta = 119.74^\circ$ | Area = 7.000 | Work (a as force, b as displacement) = -4.000

MODULE III Integral Calculus

15 hours (L:11 + T:4)

CO3

Bloom: Apply

Learning Goals

- Understand integration as reverse of differentiation
- Recall standard integrals
- Apply substitution and integration by parts
- Evaluate definite integrals
- Find area bounded by a curve and x-axis

Key Facts

- $\int x^n dx = x^{n+1}/(n+1) + C$ ($n \neq -1$)
- $\int e^x dx = e^x + C$
- $\int 1/x dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$

Integration as Reverse of Differentiation

Integration is the inverse process of differentiation. If $d/dx [F(x)] = f(x)$, then $\int f(x) dx = F(x) + C$, where C is the constant of integration.

$$\int f(x) dx = F(x) + C \quad \cdot \quad d/dx [F(x)] = f(x)$$

Example: $d/dx (x^2) = 2x \Rightarrow \int 2x dx = x^2 + C.$



Tip: The constant C accounts for all antiderivatives. Always include it in indefinite integrals.

Standard Integrals

$$\int x^n dx = x^{n+1}/(n+1) + C$$

$n \neq -1$

$$\int 1/x dx = \ln|x| + C$$

$x \neq 0$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

Rules of Integration & Substitution

Rules:

- $\int k \cdot f(x) \, dx = k \cdot \int f(x) \, dx$
- $\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$

Substitution:

- $\int f(ax+b) \, dx = (1/a) F(ax+b) + C$
- $\int f(x) \cdot f'(x) \, dx = [f(x)]^2/2 + C$
- $\int f'(x)/f(x) \, dx = \ln|f(x)| + C$

Example (substitution): $\int (2x+3)^5 \, dx$. Let $u = 2x+3$, $du = 2dx \Rightarrow dx = du/2$.
 $= 1/2 \int u^5 \, du = u^6/12 + C = (2x+3)^6/12 + C$.

Example (f'/f): $\int (2x)/(x^2+1) \, dx = \ln|x^2+1| + C$.

Integration by Parts

For product of two functions:

$$\int u \, dv = uv - \int v \, du$$

LIATE rule (choose u as the function that comes first): Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

Example: $\int x \sin x \, dx$. Let $u = x$ (algebraic), $dv = \sin x \, dx \Rightarrow du = dx$, $v = -\cos x$.
 $= x(-\cos x) - \int (-\cos x) \, dx = -x \cos x + \sin x + C$.

 **Common mistake:** Choosing the wrong u can make the integral more complex. Use LIATE.

Definite Integrals

A **definite integral** has limits of integration:

$$\int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a)$$

Example: $\int_0^1 x^2 \, dx = [x^3/3]_0^1 = 1/3 - 0 = 1/3.$

Area Bounded by a Curve and the x-axis

The area enclosed by the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is:

$$\text{Area} = \int_a^b f(x) \, dx \quad \cdot \quad (\text{if } f(x) \geq 0 \text{ on } [a,b])$$

Example: Area under $y = x^2$ from $x=0$ to $x=2$ is $\int_0^2 x^2 \, dx = [x^3/3]_0^2 = 8/3$ sq. units.

 **Engineering Application:** Definite integrals compute areas, volumes, work done, centre of mass, and signal energy in electrical engineering.

Definite Integral Calculator (polynomial)

Compute $\int_a^b x^n \, dx$. Enter a, b, and n.

a (lower limit)

0

b (upper limit)

2

n (exponent)

2

$$\int_a^b x^n \, dx = 2.666667 \quad (a=0, b=2, n=2)$$

MODULE IV Differential Equations & Laplace Transforms

16 hours (L:12 + T:4)

CO4

Bloom: Apply

Learning Goals

- Identify order and degree of differential equations
- Solve first-order variable separable and linear D.E.
- Understand Laplace transforms of elementary functions
- Apply shifting property
- Evaluate inverse Laplace transforms

Key Facts

- $L\{1\} = 1/s$
- $L\{t^n\} = n!/s^{n+1}$
- $L\{e^{at}\} = 1/(s-a)$
- $L\{\sin at\} = a/(s^2+a^2)$
- $L\{\cos at\} = s/(s^2+a^2)$

Differential Equations — Order and Degree

A **differential equation** is an equation involving derivatives of a function.

- **Order:** The highest derivative present.
- **Degree:** The exponent of the highest derivative (after removing radicals and fractions).

Example: $d^2y/dx^2 + 3(dy/dx) + 2y = 0 \rightarrow$ Order 2, Degree 1.

Example: $(d^2y/dx^2)^3 + (dy/dx)^2 + y = 0 \rightarrow$ Order 2, Degree 3.

 **Common mistake:** Degree is only defined when the equation is polynomial in derivatives. If derivatives appear in denominators or under radicals, rationalise first.

First-Order Differential Equations — Variable Separable

Form: $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x) dx$

Example: $dy/dx = xy \Rightarrow dy/y = x dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = C e^{(x^2/2)}$.

 **Tip:** Separate variables so that all y terms are on one side and all x terms on the other.

First-Order Linear Differential Equations

Form: $dy/dx + P(x)y = Q(x)$

Integrating factor: $I.F. = e^{\int P dx}$ · $y \cdot I.F. = \int Q \cdot I.F. dx + C$

Example: $dy/dx + 2y = x$. $P = 2$, $I.F. = e^{\int 2 dx} = e^{(2x)}$. $y e^{(2x)} = \int x e^{(2x)} dx + C$. Using integration by parts: $\int x e^{(2x)} dx = (x/2)e^{(2x)} - (1/4)e^{(2x)}$. So $y = (x/2) - (1/4) + C e^{(-2x)}$.

Introduction to Laplace Transforms

The **Laplace transform** of a function $f(t)$ is defined as:

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt = F(s)$$

It transforms a function of time t into a function of complex frequency s .

Malayalam support: Laplace transform സമയ domain-ൽ സ-
domain (frequency domain) ലെക്ക് മാറ്റം വരുത്തുന്നു. ഈ
differential equations-ൽ algebraic equations-ലേക്ക് മാറ്റം വരുത്തുന്നു.

 **Engineering Application:** Laplace transforms are fundamental in control systems, circuit analysis, and signal processing.

Laplace Transforms of Elementary Functions

$$L\{1\} = 1/s$$

$$(s > 0)$$

$$L\{t^n\} = n! / s^{n+1}$$

$$n = 0, 1, 2, \dots$$

$$L\{e^{at}\} = 1/(s-a)$$

$$(s > a)$$

$$L\{\sin at\} = a/(s^2+a^2)$$

$$L\{\cos at\} = s/(s^2+a^2)$$

$$L\{\sinh at\} = a/(s^2-a^2)$$

$$L\{\cosh at\} = s/(s^2-a^2)$$

Linearity property: $L\{a f(t) + b g(t)\} = a L\{f(t)\} + b L\{g(t)\}$

Example: $L\{3 + 2e^{4t}\} = 3/s + 2/(s-4).$

Shifting Property (First Shifting Theorem)

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Example: $\mathcal{L}\{e^{3t} \sin 2t\} = 2 / ((s-3)^2 + 4).$

Example: $\mathcal{L}\{e^{-2t} \cos 5t\} = (s+2) / ((s+2)^2 + 25).$

⚠ Common mistake: The shifting property shifts s by $-a$ for $\mathcal{L}\{e^{at} f(t)\}$. Remember the sign:
 $e^{at} \rightarrow F(s-a).$

Inverse Laplace Transforms

The inverse Laplace transform recovers $f(t)$ from $F(s)$. Common inverses:

$$\mathcal{L}^{-1}\{1/s\} = 1$$

$$\mathcal{L}^{-1}\{1/s^2\} = t$$

$$\mathcal{L}^{-1}\{1/(s-a)\} = e^{at}$$

$$\mathcal{L}^{-1}\{1/(s^2+a^2)\} = (1/a) \sin at$$

$$\mathcal{L}^{-1}\{s/(s^2+a^2)\} = \cos at$$

Example: $\mathcal{L}^{-1}\{1/(s^2+9)\} = (1/3) \sin 3t.$

Example: $\mathcal{L}^{-1}\{s/(s^2+16)\} = \cos 4t.$



Tip: Use partial fractions to simplify $F(s)$ before finding inverse transforms.



Laplace Transform Helper

Select a function to see its Laplace transform and inverse.

Function $f(t)$

1

a

2

b (if needed)

3

$$\mathcal{L}\{1\} = 1/s \mid \mathcal{L}^{-1}\{1/s\} = 1$$



Formula Bank

$$z = a + ib, i^2 = -1$$

Complex number

$$|z| = \sqrt{a^2 + b^2}$$

Modulus

$$\theta = \tan^{-1}(b/a)$$

Argument

$$z = r(\cos \theta + i \sin \theta)$$

Polar form

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

Dot product

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \sin \theta \cdot \hat{n}$$

Cross product

$$\int x^n dx = x^{n+1}/(n+1) + C$$

Power rule

$$\int u dv = uv - \int v du$$

Integration by parts

$$\int_a^b f(x) dx = F(b) - F(a)$$

Definite integral

$$I.F. = e^{\int P dx}$$

Linear D.E.

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Laplace transform

$$L\{e^{at} f(t)\} = F(s-a)$$

Shifting property



Diagram Library for Rapid Revision



Argand Plane

Module 1 — Argand Plane showing complex number representation.



Vector Geometry

Visualise dot and cross products in 3D space. Module 2



Area Under Curve

Visualise area bounded by a curve and x-axis. Module 3



Self-Learning Activities (Official)

Week 1-4 · 16 h

Module I — Complex Numbers

- Find conjugate, modulus, amplitude, polar form
- Problems on equality, addition, subtraction
- Multiply and divide complex numbers
- Applications in engineering areas

Week 5-8 · 15 h

Module II — Vector Algebra

- Identify scalars and vectors
- Vector addition, subtraction, scalar multiplication
- Dot product and cross product problems
- Work done and moment problems
- Applications of vectors

Week 9-12 · 14 h

Module III — Integral Calculus

- Find integrals of functions
- Evaluate definite integrals
- Area bounded by curve and x-axis
- Applications of integral calculus

Week 13-15 · 15 h

Module IV — D.E. & Laplace

- Solve first-order differential equations
- Problems using first shifting property
- Evaluate Laplace transforms of functions
- Applications of D.E. and Laplace

Total self-learning hours: 60 (as per syllabus)



Assessment Methodology

CIE / ESE Breakup

Mode	Attendance	CA1	CA2	CA3	CA4	Series Exam (2 modules)	Series Exam (2 modules)	ESE
Duration	—	2h	2h	2h	2h	2h	2h	2.5h
Exam mark	—	15	15	15	15	15	15	60
Converted to	5	15				20		60
Total	40 (CIE) + 60 (ESE)							

Series Examination Pattern

Duration: 2 hours

Total marks: 50 (converted to 20)

- Part A: 2×1 mark
- Part B: 2×3 marks
- Part C: 2×7 marks (with choice)

End Semester Examination

Duration: 2.5 hours

Total marks: 60

- Part A: 1 mark $\times$ 12 questions (2 per module)
- Part B: 3 marks $\times$ 6 questions (2 out of 3 per module)
- Part C: 7 marks $\times$ 6 questions (1 set with choice per module)

? Module-wise Questions with Answers

Module I — Complex Numbers

Q1: Find the modulus and argument of $z = 1 + i$.▼

$$|z| = \sqrt{(1^2+1^2)} = \sqrt{2}. \theta = \tan^{-1}(1/1) = 45^\circ.$$

Q2: Express $z = 2 - 2i$ in polar form.▼

$$r = \sqrt{(4+4)} = 2\sqrt{2}. \theta = -45^\circ. z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ)).$$

Q3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.▼

$$(1+2i)(3-i) = 3-i+6i-2i^2 = 3+5i+2 = 5+5i.$$

Module II — Vector Algebra

Q1: Find the magnitude of $\mathbf{v} = 3\hat{i} + 4\hat{j}$.▼

$$|\mathbf{v}| = \sqrt{9+16} = 5.$$

Q2: Compute the dot product of $\mathbf{a} = (2,3)$ and $\mathbf{b} = (1,-2)$.▼

$$\mathbf{a} \cdot \mathbf{b} = 2(1)+3(-2) = 2-6 = -4.$$

Q3: Find the cross product of $\mathbf{a} = 2\hat{i}+3\hat{j}$ and $\mathbf{b} = 1\hat{i}-2\hat{j}$.▼

$$\mathbf{a} \times \mathbf{b} = (3 \cdot 0 - 0 \cdot (-2))\hat{i} - (2 \cdot 0 - 0 \cdot 1)\hat{j} + (2 \cdot (-2) - 3 \cdot 1)\hat{k} = -7\hat{k}.$$

Q4: What is the area of a parallelogram formed by vectors $\mathbf{a}$ and $\mathbf{b}$ from Q3?▼

$$\text{Area} = |\mathbf{a} \times \mathbf{b}| = 7 \text{ sq. units.}$$

Module III — Integral Calculus

Q1: Evaluate $\int 3x^2 \, dx$.▼

$$\int 3x^2 \, dx = 3 \cdot (x^3/3) + C = x^3 + C.$$

Q2: Evaluate $\int_0^1 x \, dx$.▼

$$[x^2/2]_0^1 = 1/2 - 0 = 1/2.$$

Q3: Find the area under $y = x^2$ from $x=0$ to $x=2$.▼

$$\int_0^2 x^2 \, dx = [x^3/3]_0^2 = 8/3 \text{ sq. units.}$$

Module IV — Differential Equations & Laplace

Q1: Find the order and degree of $\frac{d^2y}{dx^2} + 3(\frac{dy}{dx}) + 2y = 0$.▼

Order = 2, Degree = 1.

Q2: Solve $\frac{dy}{dx} = 2xy$.▼

$\frac{dy}{y} = 2x \, dx \Rightarrow \ln|y| = x^2 + C \Rightarrow y = C e^{(x^2)}.$

Q3: Find $L\{\sin 3t\}$.▼

$L\{\sin 3t\} = 3/(s^2+9).$

Q4: Find $L^{-1}\{1/(s^2+25)\}$.▼

$L^{-1}\{1/(s^2+25)\} = (1/5) \sin 5t.$



Practice Model Paper

Note: This is a practice paper based on the official pattern, not an official SITTTTR model question paper.

Course 2001B — Mathematics for Electrical Sciences

Duration: 2.5 hours · **Total marks:** 60

All modules are covered. Answer all parts.

Part A (1 mark each — 12 questions, 2 per module)

- **Q1.** What is the conjugate of $2 + 5i$? (M1)
- **Q2.** Find $|3 - 4i|$. (M1)
- **Q3.** What is a unit vector? (M2)
- **Q4.** When are two vectors perpendicular? (M2)
- **Q5.** $\int 2x \, dx = ?$ (M3)
- **Q6.** What is the order of $\frac{d^2y}{dx^2} + y = 0$? (M3)
- **Q7.** Define Laplace transform. (M4)
- **Q8.** Find $L\{1\}$. (M4)
- **Q9-12.** (Additional questions covering all modules)

Part B (3 marks each — 6 questions, 2 out of 3 per module)

- **M1:** Express $1 - i$ in polar form. *OR* Find modulus and argument of $-1 + i$.
- **M2:** Given $a = (1, 2, 0)$, $b = (0, 1, 1)$, find $a \cdot b$ and $a \times b$. *OR* Find work done if $F = 2\hat{i} + 3\hat{j}$ and $d = 1\hat{i} - 2\hat{j}$.
- **M3:** Evaluate $\int x \sin x \, dx$. *OR* Find area under $y = 2x$ from $x=0$ to $x=3$.
- **M4:** Solve $dy/dx + 2y = x$. *OR* Find $L\{e^{3t} \sin 2t\}$.

Part C (7 marks each — 6 questions, 1 set with choice per module)

- **M1:** If $z_1 = 2 + 3i$ and $z_2 = 1 - 2i$, find $z_1 + z_2$, $z_1 - z_2$, $z_1 \cdot z_2$, and z_1/z_2 . *OR* Express $z = 2 + 2i$ in polar form and find z^3 .
- **M2:** For vectors $a = 2\hat{i} + \hat{j} - \hat{k}$ and $b = \hat{i} - 2\hat{j} + 3\hat{k}$, find $a \cdot b$, $a \times b$, area of parallelogram, and angle between them. *OR* Find moment $\tau = r \times F$ for $r = 3\hat{i} + 2\hat{j}$, $F = 1\hat{i} - 4\hat{j}$.
- **M3:** Evaluate $\int_0^2 (x^2 + 2x) \, dx$. Find area bounded by $y = x^2$ and x -axis from $x=0$ to $x=2$. *OR* Evaluate $\int x^2 e^x \, dx$.
- **M4:** Solve $dy/dx + 2y = e^x$. Find $L\{e^{-2t} \cos 3t\}$. *OR* Find $L^{-1}\{1/(s^2 + 4s + 13)\}$ (using completing square and shifting).

Quick Revision

Module I — Complex Numbers

- $z = a + ib, i^2 = -1$
- $|z| = \sqrt{a^2+b^2}, \theta = \tan^{-1}(b/a)$
- Polar: $z = r(\cos \theta + i \sin \theta)$
- Operations: add real/imag separately

Module II — Vector Algebra

- $a \cdot b = |a||b| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$
- Area = $|a \times b|$, Work = $F \cdot d$, Moment = $r \times F$

Module III — Integral Calculus

- $\int x^n dx = x^{n+1}/(n+1) + C$
- $\int u dv = uv - \int v du$
- Definite: $\int_a^b = F(b) - F(a)$
- Area = $\int_a^b f(x) dx$

Module IV — D.E. & Laplace

- I.F. = $e^{\int P dx}$ for linear D.E.
- $L\{1\} = 1/s, L\{e^{at}\} = 1/(s-a)$
- $L\{\sin at\} = a/(s^2+a^2), L\{\cos at\} = s/(s^2+a^2)$
- $L\{e^{at}f(t)\} = F(s-a)$

⚠ Common Mistakes — Exam Checklist

- ✓ In complex division, multiply by conjugate of denominator
- ✓ In vectors, cross product gives a vector, dot product gives a scalar
- ✓ In integration, always add +C for indefinite integrals
- ✓ In Laplace, shifting property: $L\{e^{at}f(t)\} = F(s-a)$ (note the sign)
- ✓ In differential equations, identify order and degree correctly
- ✓ In inverse Laplace, use partial fractions when needed



Glossary

Term	Meaning
Complex number	$z = a + ib$, where $i^2 = -1$
Modulus	$ z = \sqrt{a^2+b^2}$, distance from origin
Argument	$\theta = \tan^{-1}(b/a)$, angle with real axis
Polar form	$z = r(\cos \theta + i \sin \theta)$
Scalar	Quantity with only magnitude
Vector	Quantity with magnitude and direction
Dot product	Scalar product: $a \cdot b = a b \cos \theta$
Cross product	Vector product: $a \times b = a b \sin \theta \cdot \hat{n}$
Integration	Reverse process of differentiation
Definite integral	$\int_a^b f(x) \, dx = F(b) - F(a)$
Differential equation	Equation involving derivatives
Laplace transform	$L\{f(t)\} = \int_0^\infty e^{-st} f(t) \, dt$
Shifting property	$L\{e^{at}f(t)\} = F(s-a)$
Inverse Laplace	Recovers $f(t)$ from $F(s)$
Integrating factor	$I.F. = e^{\int P \, dx}$ for linear D.E.



Learning Resources

Textbooks		
Sl. No.	Title	Author / Publication
1	Mathematics for Electrical Sciences	SITTTR Kalamasser

References		
Sl. No.	Title	Author / Publication
1	Schaum's outline of Complex Variables	Murray R. Spiegel, Seymour Lipschutz, John J. Schiller, McGraw Hill
2	Schaum's outline of Vector Analysis	Murray R. Spiegel, McGraw Hill
3	Advanced Engineering Mathematics (10th Ed.)	Erwin Kreyszig, Wiley & Sons
4	A Textbook of Engineering Mathematics	N.P. Bali, Manish Goyal, University Science Press
5	Advanced Engineering Mathematics	B. S. Grewal, Mercury Learning & Information

Web resources: None officially listed in the syllabus.